



Safe Scroll

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1. Introduction

The purpose of this document is to outline the software requirements for a Chrome browser extension designed to detect and hide hateful memes on social media platforms. This extension will serve as a tool for users to filter out harmful visual content, enhancing their online experience by promoting a safer and more positive environment on popular social media sites.

2. Vision Document

2.1. Problem Statement

Table 1 Problem Statement

The Problem	The presence of hateful memes on social media platforms that spread toxic, discriminatory, and offensive content.
Affects	Social media users, including individuals, families, and organizations who are exposed to harmful visual content.
The Impact	An increase in online harassment, negative mental health effects, and a decline in the overall quality of online interactions. This problem also poses a risk for businesses and individuals concerned about their public image and mental well-being.
Successful Solution	A Chrome extension that detects and automatically hides hateful memes on popular social media platforms, providing users with a safer and more enjoyable browsing experience.

2.2. Business Opportunity

With the rising emphasis on creating safe online spaces, there is a significant opportunity to develop a Chrome extension that hides hateful memes. This tool will cater to both individual users and businesses that seek to maintain positive brand reputations by avoiding exposure to hateful content. By capitalizing on this market, we can provide a solution that helps enhance online user engagement and improves mental well-being by reducing exposure to negative content.

2.3. Objectives

- To develop a Chrome extension that identifies and hides hateful memes on social media platforms.
- To integrate an advanced meme detection model for filtering harmful visual content.

- To create a seamless and user-friendly experience that automatically detects and blocks offensive memes in real-time.
- To ensure the extension is adaptable to multiple social media platforms, including Facebook, Twitter, and Instagram.

2.4. Scope

The scope of this project includes the design, development, and deployment of a Chrome extension that integrates a state-of-the-art hateful meme detection model. The extension will focus on detecting and hiding memes with offensive or hateful content across various social media platforms.

2.5. Constraints

- The effectiveness of meme detection is dependent on the accuracy and performance of the underlying model.
- The extension must operate within the constraints of the Chrome Web Store's policies.
- Integration with social media platforms may be limited by the availability of APIs or browser access controls.

2.6. Stakeholder and User Descriptions

2.6.1. Market Demographics

- Social media users who are concerned about online toxicity and prefer a safer browsing experience.
- Parents or guardians looking to protect children from exposure to harmful content.

2.6.2. Stakeholder Summary

Table 2 Stakeholder summary

Name	Description	Responsibilities
Users	Individuals who want to filter hateful memes while browsing social media..	Use the extension to block harmful content.
Development Team	Developers responsible for building the extension.	1. Design, build, and test the extension. 2. Integrate hateful meme detection models into the system.
Supervisor	Provides guidance, oversees the project's progress, and ensures alignment with overall goals.	Monitoring development, providing feedback, approving milestones, and ensuring timely delivery.

2.6.3. User Environment

Users will access the Chrome extension by downloading it from the Chrome Web Store. Once installed, the extension will integrate with popular social media platforms (e.g., Facebook, Twitter, Instagram) through the Chrome browser, automatically detecting and hiding hateful memes.

2.6.4. Stakeholder Profiles

2.6.4.1. Supervisor of the Project

Table 3 Supervisor detail

Representatives	Supervisor: Dr.Rabia Maqsood
Description	The supervisor oversees the development of the hateful meme de- tection extension, guiding the team and ensuring that the product aligns with project goals and user needs.
Type	Technical stakeholder with expertise in machine learning, and soft- ware development.
Responsibilities	1. Provides guidance to the development team. 2. Ensures the extension meets technical and user requirements. 3. Validates the performance of the hateful meme detection model. 4. Tracks project progress and ensures milestones are met on time. 5. Supports resource management and testing.
Success Criteria	Completion of the Chrome extension with core features, including meme detection, subscription, and profile management, within the defined timelines.
Involvement	1. Requirement reviewer 2. Validates the extension’s implementation and feature perfor- mance

2.6.4.2. Development Team

Table 4 Developer detail

Representatives	Wajee ul Hassan Afaq Ahmed Om Parkash
Description	The development team is responsible for the technical implementation of the extension, including the integration of hateful meme detection models, UI design, and overall functionality.
Type	Technical stakeholders with expertise in software development, machine learning, and user experience design.
Responsibilities	1. Develops the Chrome extension. 2. Integrates machine learning models for hateful meme detection. 3. Designs the UI/UX for a smooth user experience. 4. Conducts testing and debugging. 5. Ensures performance and compatibility across social media platforms.
Success Criteria	Successful deployment of the extension with seamless meme detection and hiding features, while providing users with analytics and customization options.
Involvement	1. Developers of the core extension 2. Reviewers of functionality and performance 3. Responsible for feature implementation and testing
Deliverable	1. Documentation of product 2. Fully functioning extension

2.6.4.3. Users

Table 5 User detail

Representatives	Social media users, parents/guardians
Description	Users are individuals or organizations that use the Chrome extension to block and hide hateful memes on social media platforms. This includes general users who want a safer online experience.
Type	Non-technical stakeholders.
Responsibilities	1. Install and configure the Chrome extension to suit their needs. 2. Manage subscription plans and profile settings. 3. Monitor and adjust content filtering levels. 4. View analytics on meme filtering performance.
Involvement	1. Primary users of the system 2. Provide feedback on usability and feature performance

3. System Requirements Specification

3.1. System Features

- User management (Sign up/Sign in)
- Subscription management
- Meme analytics
- Profile management

3.2. Functional Requirements

Table 5 Functional requirement detail

ID	Functional Requirements
FR-01	User Features: Users can sign up, sign in, sign out, and manage their profile within the Chrome extension..
FR-02	Subscription Service: Users can purchase subscriptions to access premium features of the meme analytics tool.
FR-03	Meme Analytics: The extension will provide insights and statistics on the types of memes hidden by the system to subscribers.
FR-04	Profile Management: Users can manage their profile, including viewing subscription status and analytics data.

3.3. Non-Functional Requirements

Table 6 Non functional requirement detail

ID	Non-Functional Requirements
NFR-01	Performance: The extension should hide the meme before user can reach it
NFR-02	Security: The platform will employ encryption for sensitive data, authentication protocols.
NFR-03	Interoperability: The system will integrate with external services such as stripe and email notification systems
NFR-04	Usability: The extension interface will be intuitive and accessible for users with basic internet knowledge.
NFR-05	Maintainability: Regular updates to the extension will be rolled out without affecting user data or service.

4. Use Case Diagram

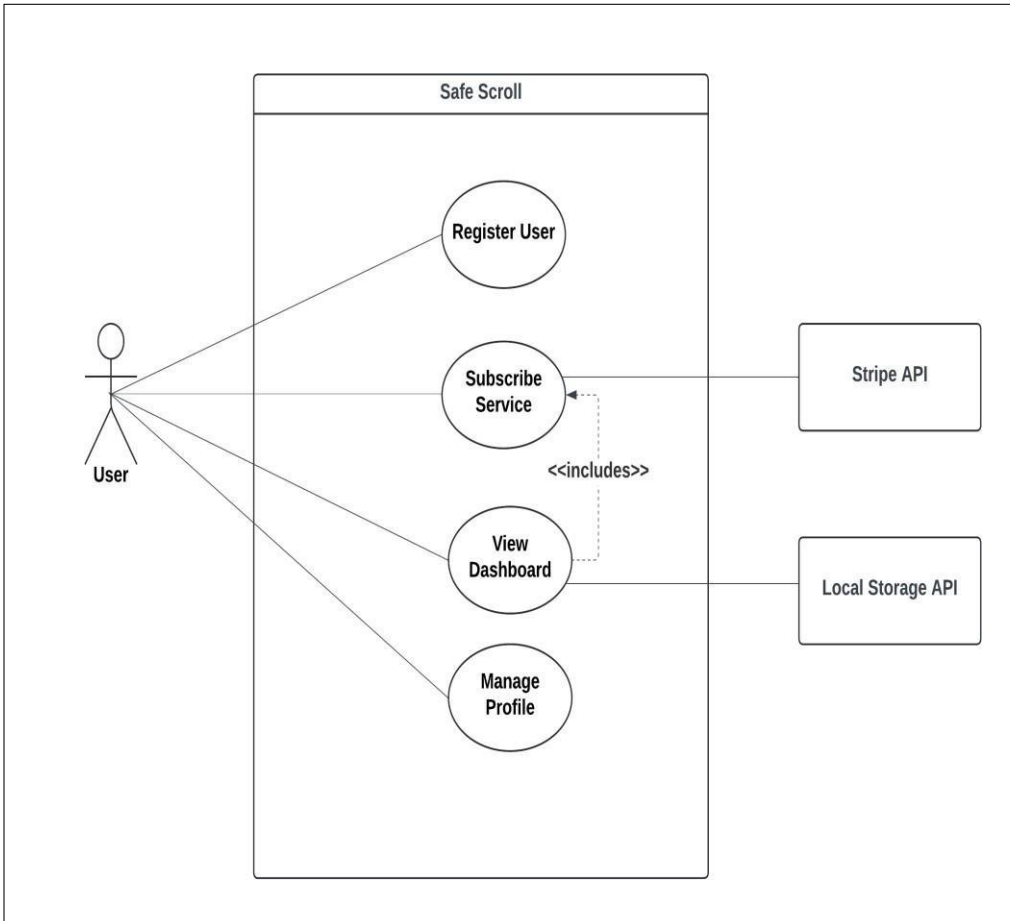


Figure 1: Use case diagram

5. High Level Use Case

Table 7 Registration (high level use case)

Use Case ID	UC-01
Use Case Name	Registration
Actor	User
Type	Primary
Description	The user can create a new account or sign in to access the platform's services.

Table 8 Subscription for more features (high level use case)

Use Case ID	UC-02
Use Case Name	Subscription
Actor	User
Type	Primary
Description	The user can subscribe to the premium service of advanced analytics.

Table 9 Profile management (high level use case)

Use Case ID	UC-03
Use Case Name	Profile Management
Actor	User
Type	Primary
Description	The user can view and update their profile, including personal information, preferences, and settings.

Table 10 Meme Analytics (high level use case)

Use Case ID	UC-04
Use Case Name	Meme Analytics
Actor	User
Type	Primary
Description	The user can access reports and analytics about the memes flagged as hateful by the system.

6. Expanded Use Case

Table 11 Registration (Expanded use case)

Use Case ID	UC-01
Use Case Name	Subscription
Actor	User
Type	Primary
Description	The user subscribes to premium services, providing access to additional features, such as detailed meme analytics and higher detection limits.
Trigger	The user navigates to the subscription page and initiates a subscription process.
Pre-condition	The user has a valid account and is logged in.
Post-condition	The user successfully subscribes and receives access to premium services.
Normal Flow	The user navigates to the subscription page. The user selects a subscription plan. The system redirects the user to a payment page integrated with the Stripe API. The user enters payment details and confirms. Upon successful payment, the system grants access to premium features.
Requirement	The system must be integrated with a payment gateway (Stripe API).
Frequency of Use	Low

Table 12 Meme Analytics (Expanded use case)

Use Case ID	UC-02
Use Case Name	Meme Analytics
Actor	User
Type	Primary
Description	The user can view detailed analytics and reports about the memes flagged by the system for containing hateful content.
Trigger	The user selects the “Meme Analytics” option from the dashboard.
Pre-condition	The user is logged into their account and has access to the analytics feature (premium feature if applicable).
Post-condition	The user is presented with detailed analytics, including meme trends, and frequency of flagged memes.
Normal Flow	The user clicks on the “Meme Analytics” section in the dashboard. The system retrieves and displays flagged memes in the analytics dashboard. The system displays the following information in the analytics dashboard: • Total count of flagged memes. • Count of memes categorized as hateful. • Count of memes categorized as safe. • The date each meme was flagged.
Alternative Flow	None
Requirement	Internet should be available
Frequency of Use	High

Table 13 Profile management (Expanded use case)

Use Case ID	UC-03
Use Case Name	Profile Management
Actor	User
Type	Primary
Description	The user can view and update their profile, including personal information, account settings, and preferences.
Trigger	The user navigates to the profile management page from the dashboard.
Pre-condition	The user is logged into the system.
Post-condition	The user's profile is successfully updated and stored in the system.
Normal Flow	The user clicks on the "Manage Profile" section. The system displays the user's profile information (e.g., name, email, subscription status). The user can edit any profile details (e.g., email, password, preferences). The user saves the changes. The system updates the user's information in the database and displays a confirmation message.
Alternate Flow	None
Requirement	The user should be registered into the application, should have internet available.
Frequency of Use	Low

7. Activity Diagram

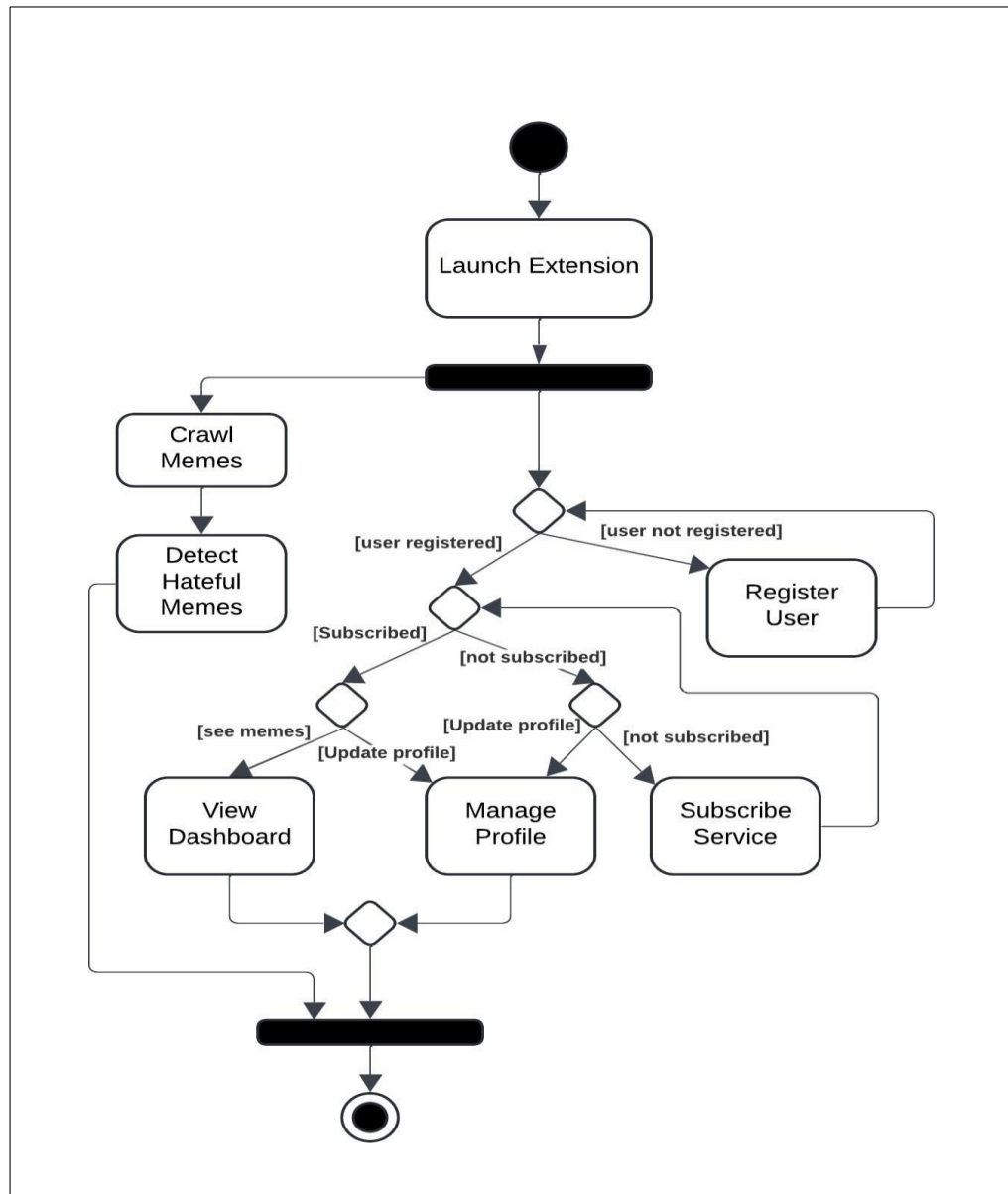


Figure 2: Activity Diagram

8. System Sequence Diagram (SSD)

8.1. Sign In/Registration

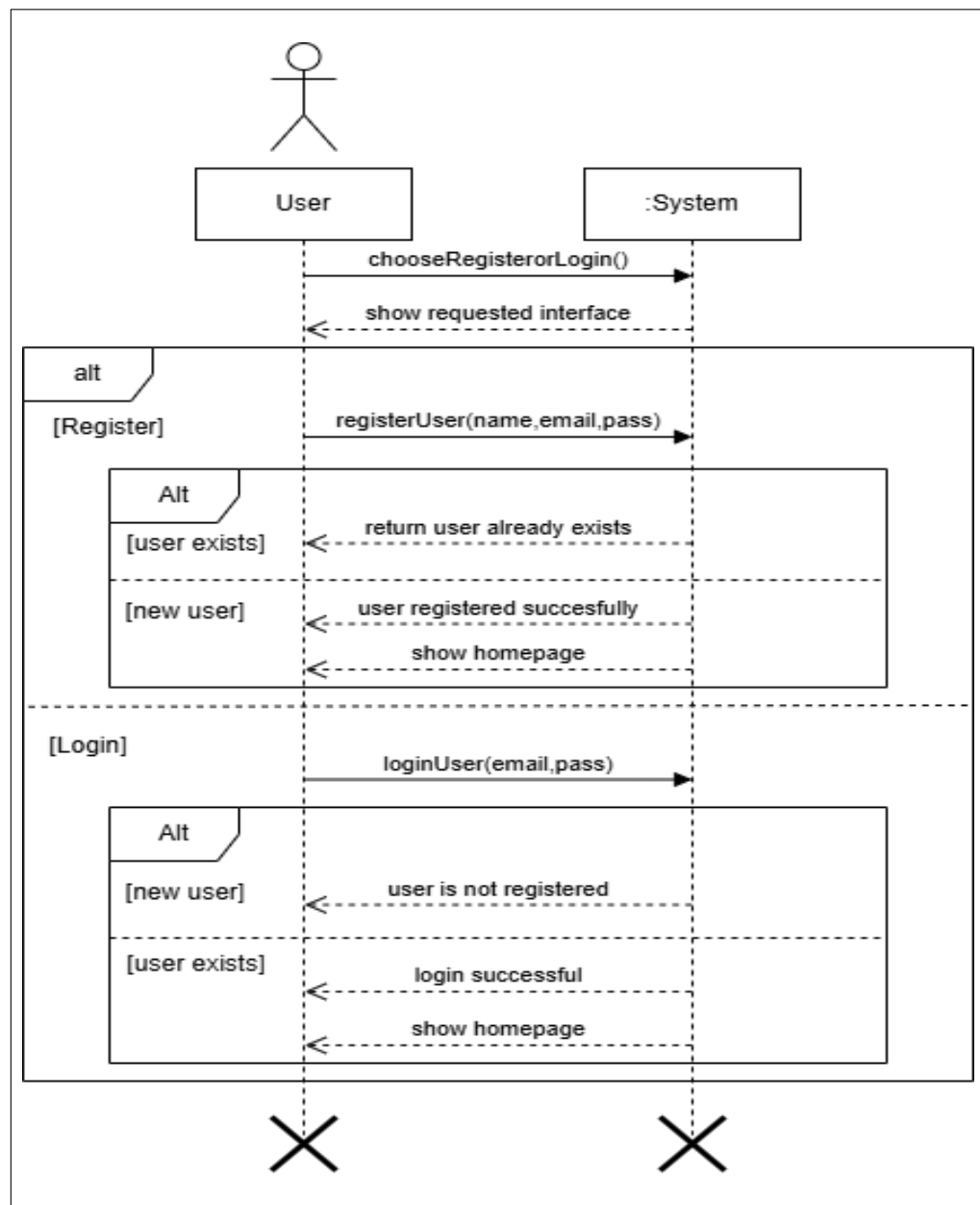


Figure 3: Sign In/Registration (SSD)

8.2. Manage Profile

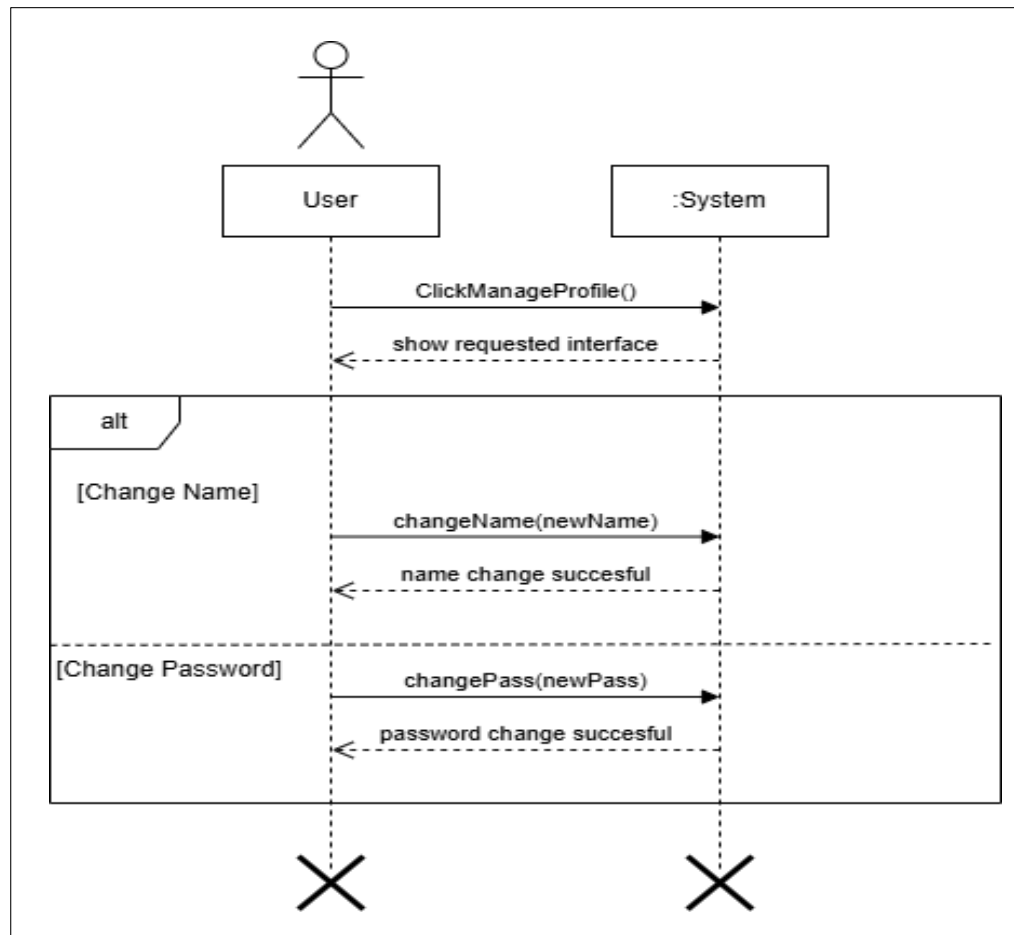


Figure 4: Manage Profile (SSD)

8.3. Buy Subscription

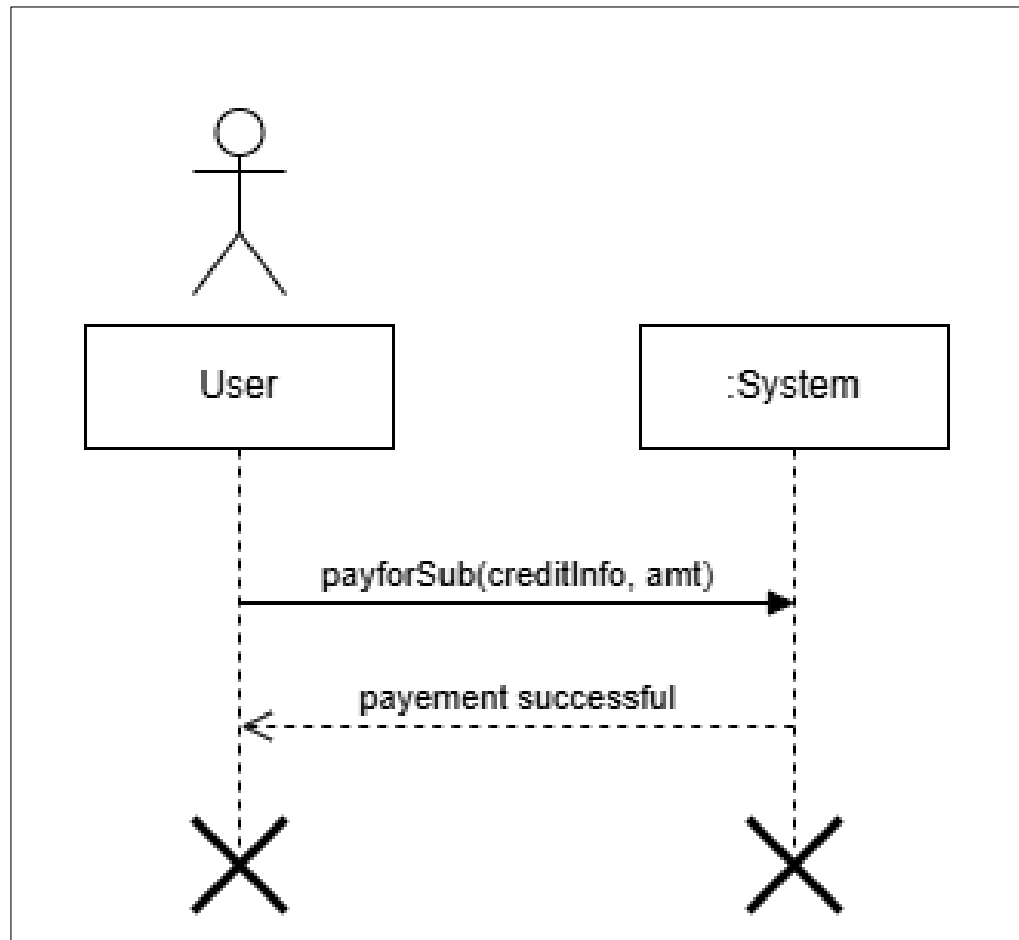


Figure 5: Buy Subscription (SSD)

8.4. Meme Analytics

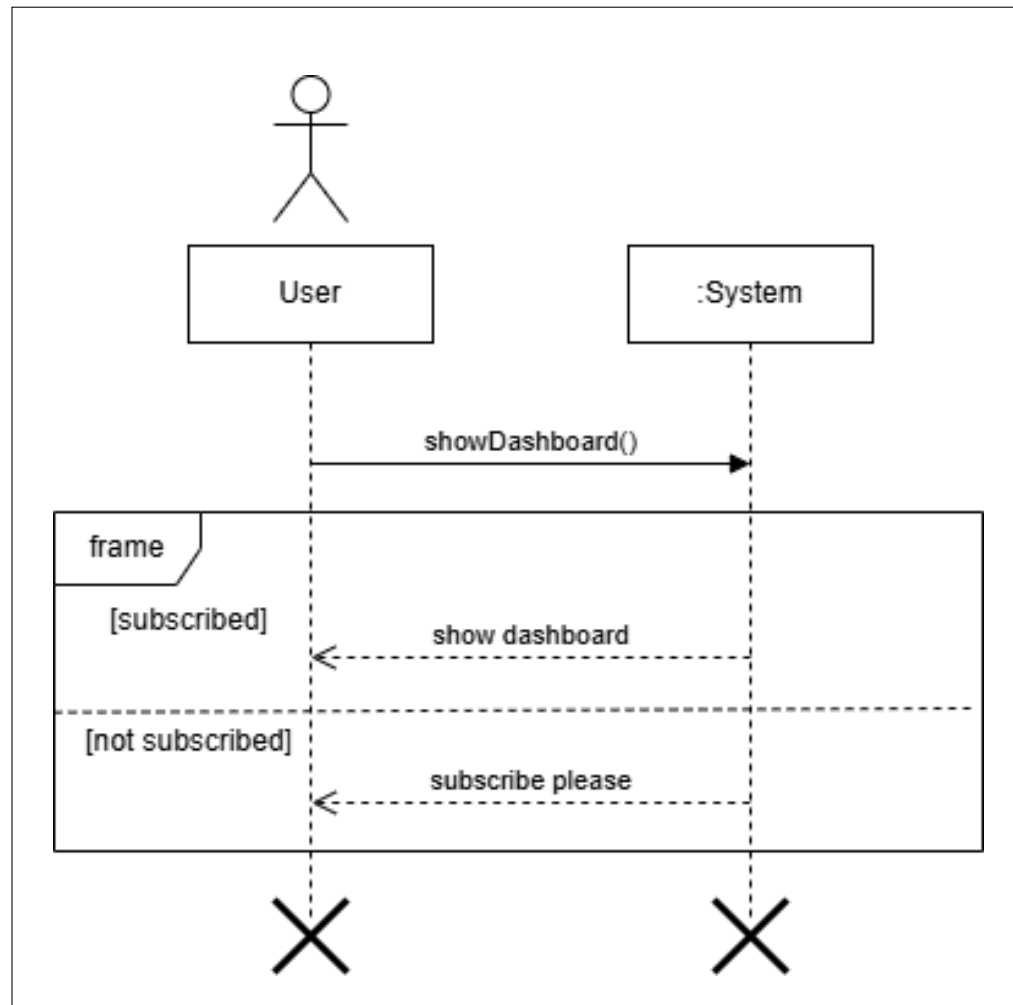


Figure 6: Meme Analytics (SSD)

8.5. Meme Detection

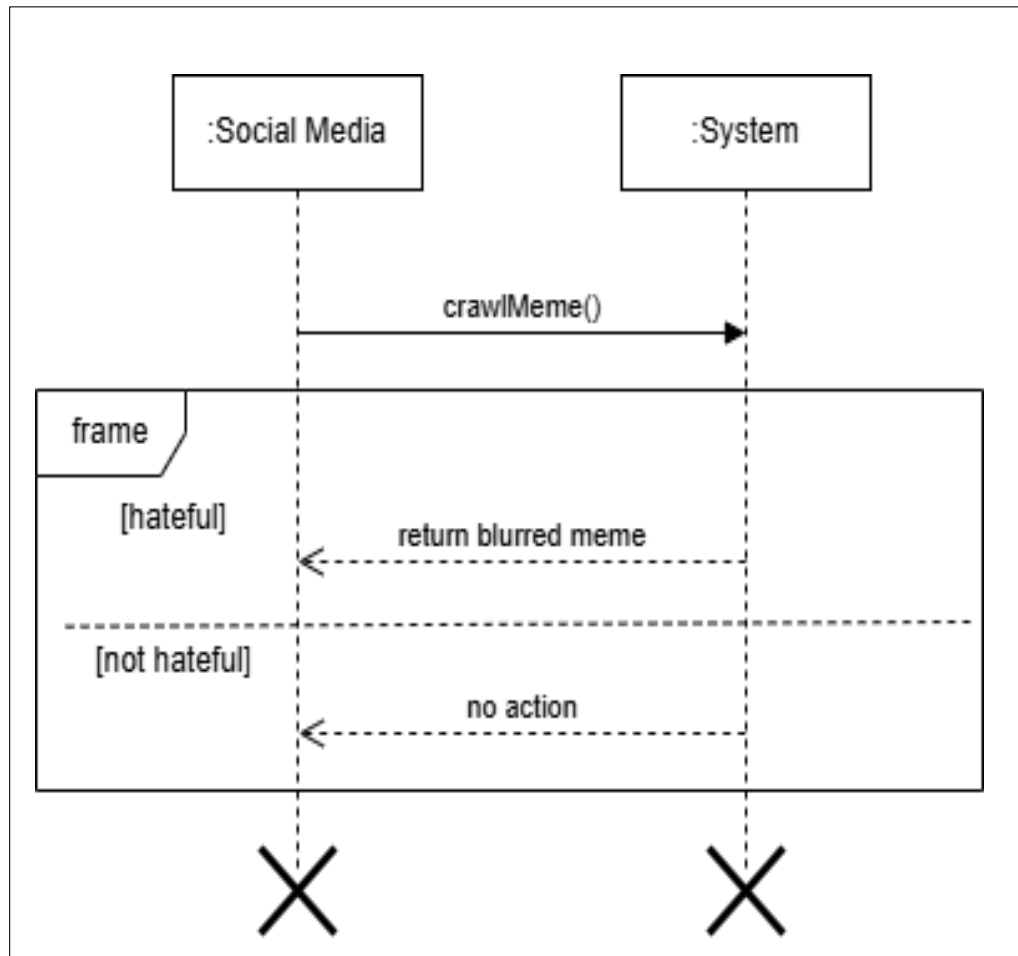


Figure 7: Meme Detection (SSD)

9. Domain Model

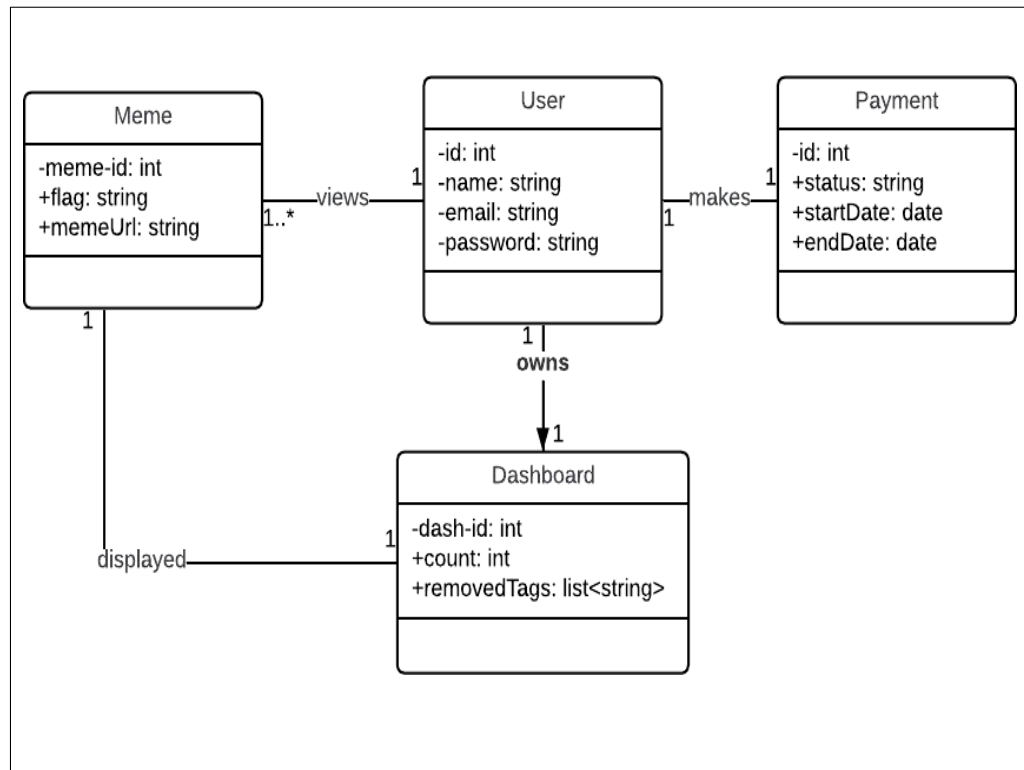


Figure 8: Domain Model

10. Component Diagram

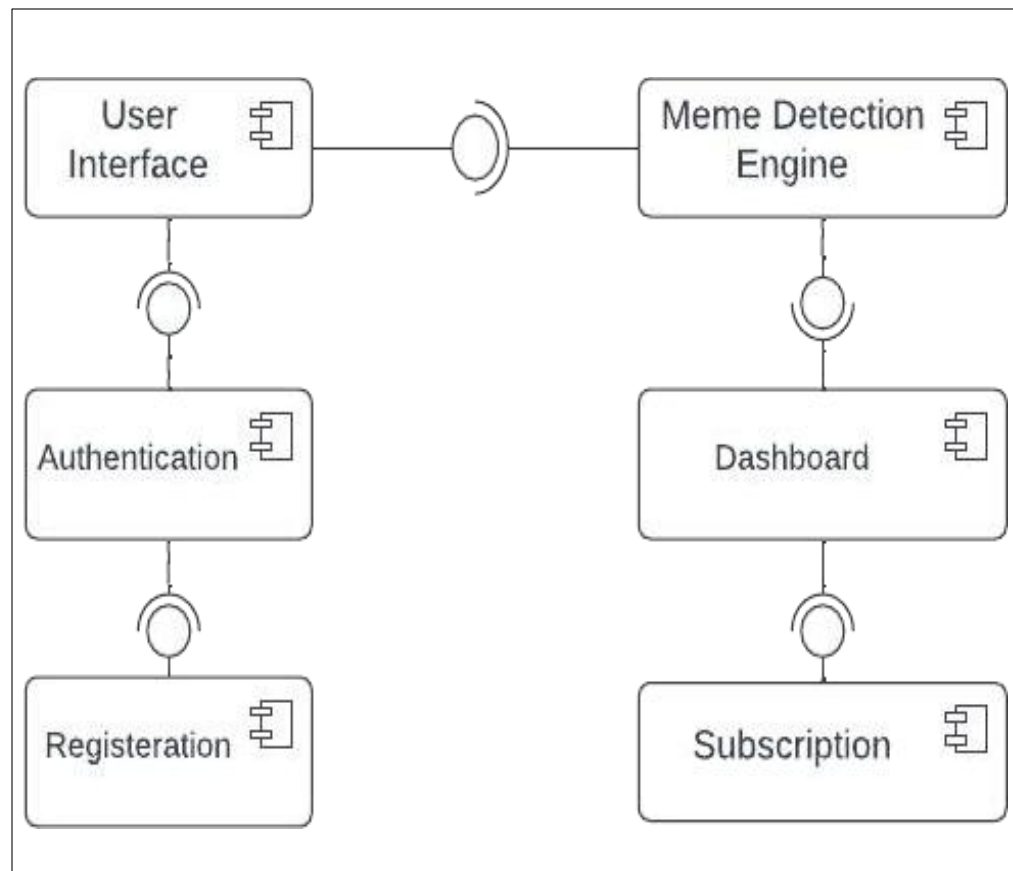


Figure 9: Component Diagram

11. Architecture Diagram

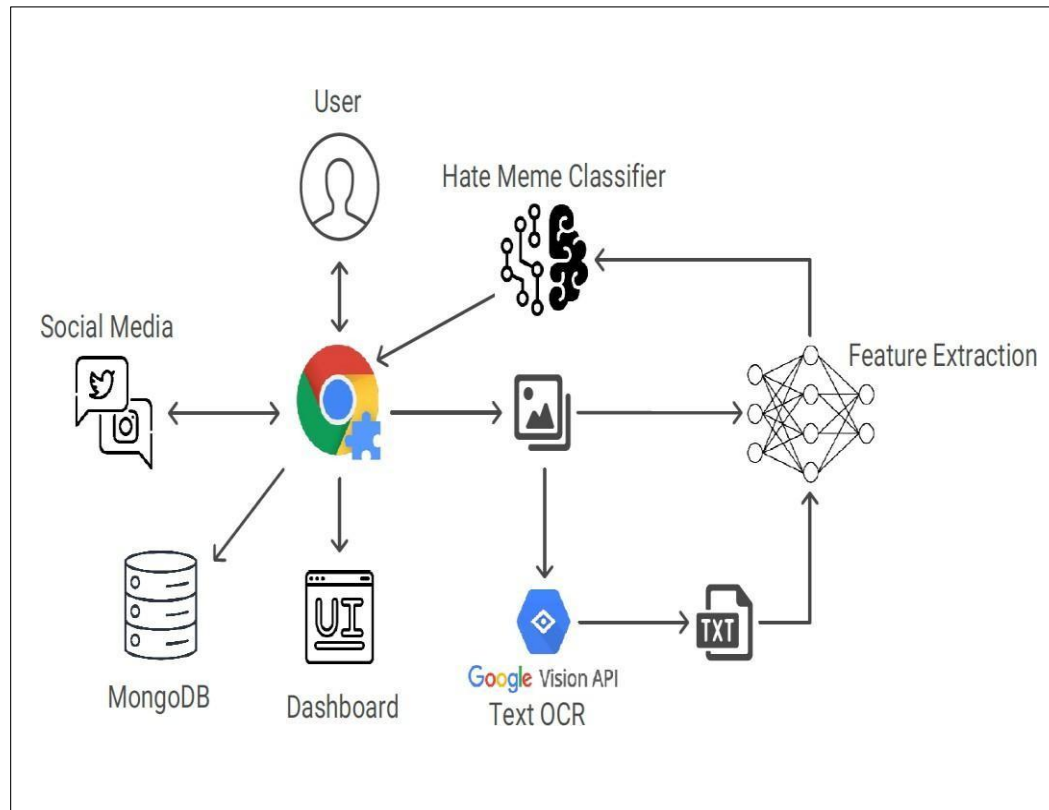


Figure 10: Architecture Diagram

12. Data Flow Diagram

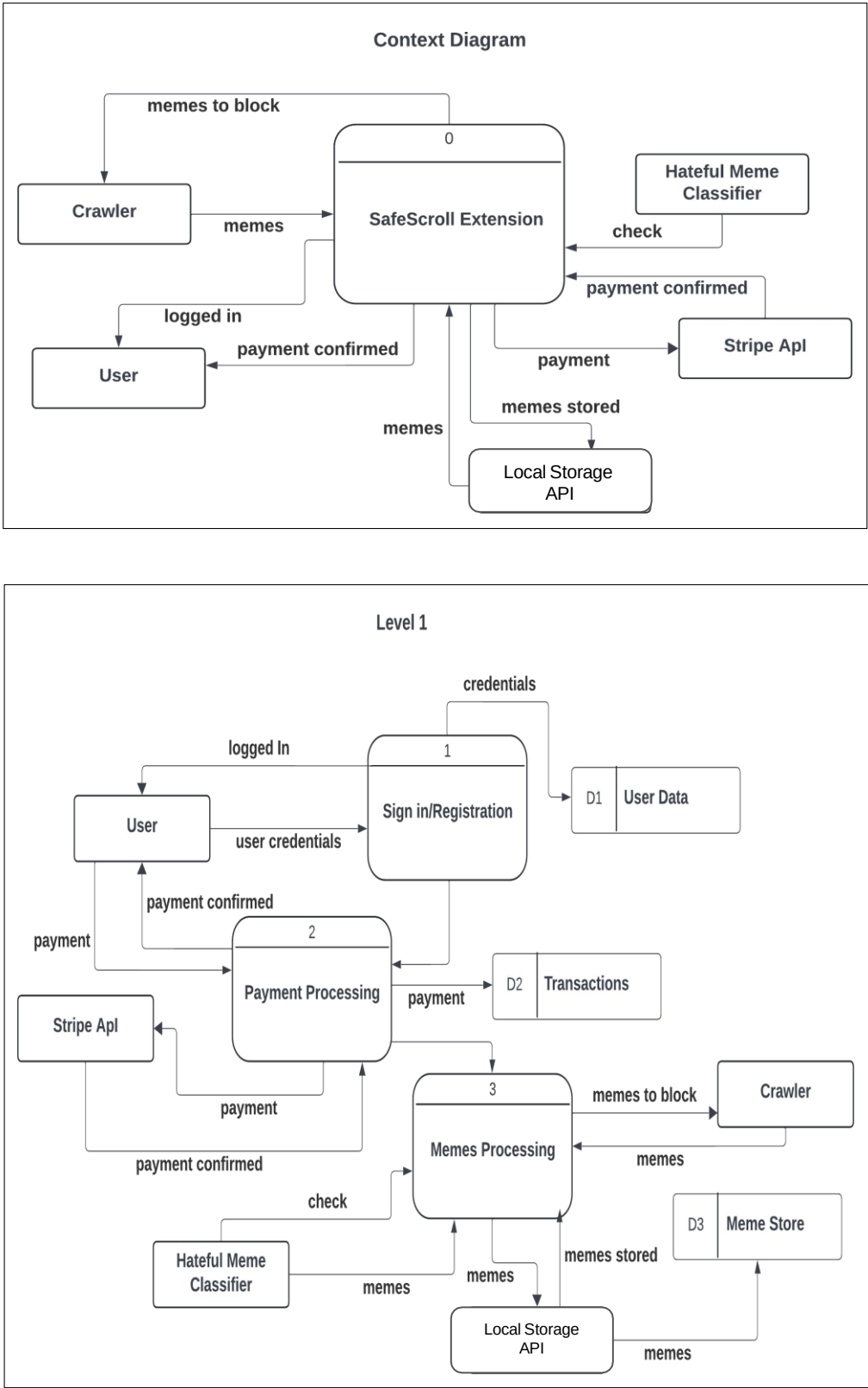


Figure 11: Data Flow Diagram

13. JSON Tree

13.1. JSON Tree Diagram 1

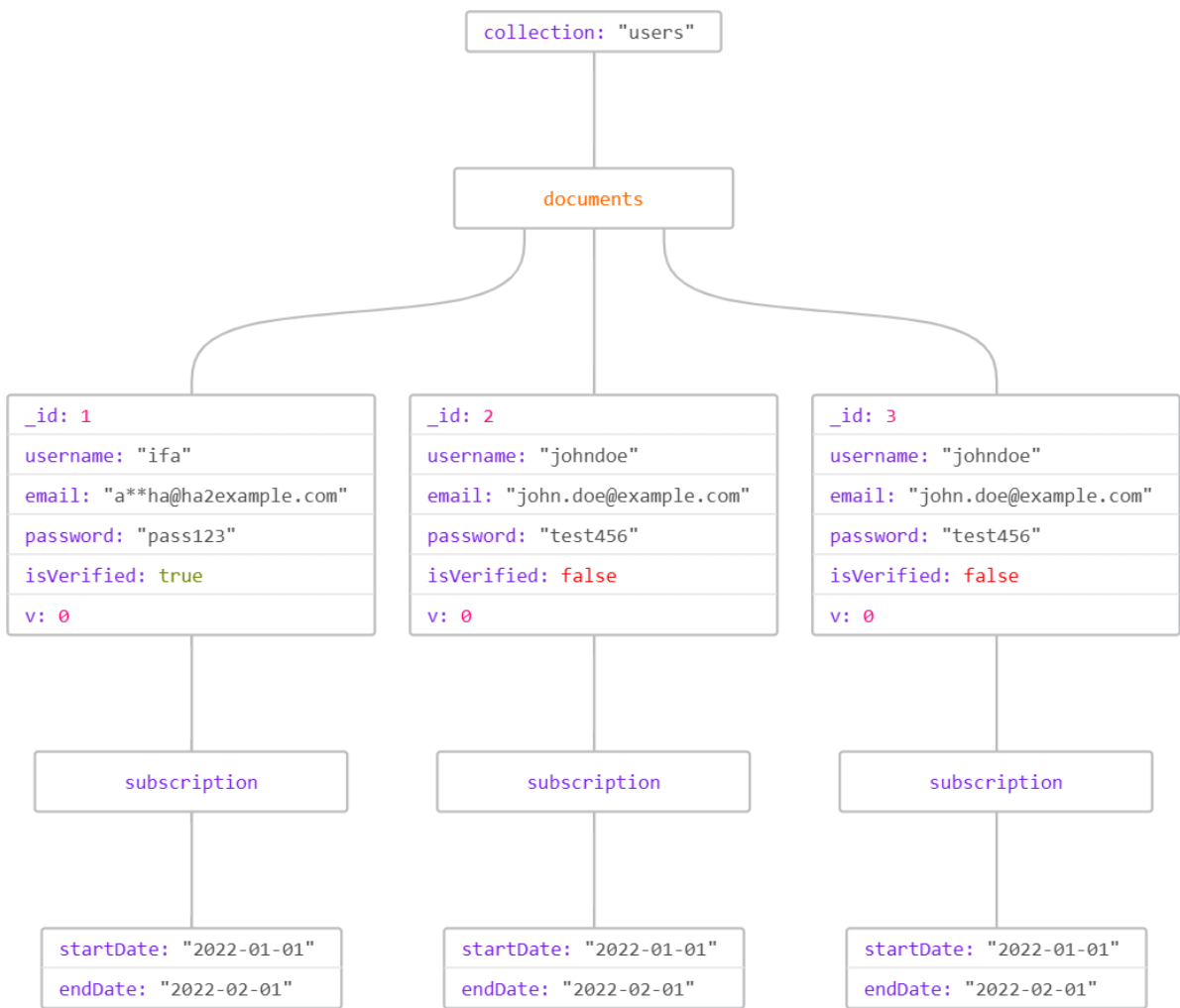


Figure 12: JSON Tree Diagram 1

14. Package Diagram

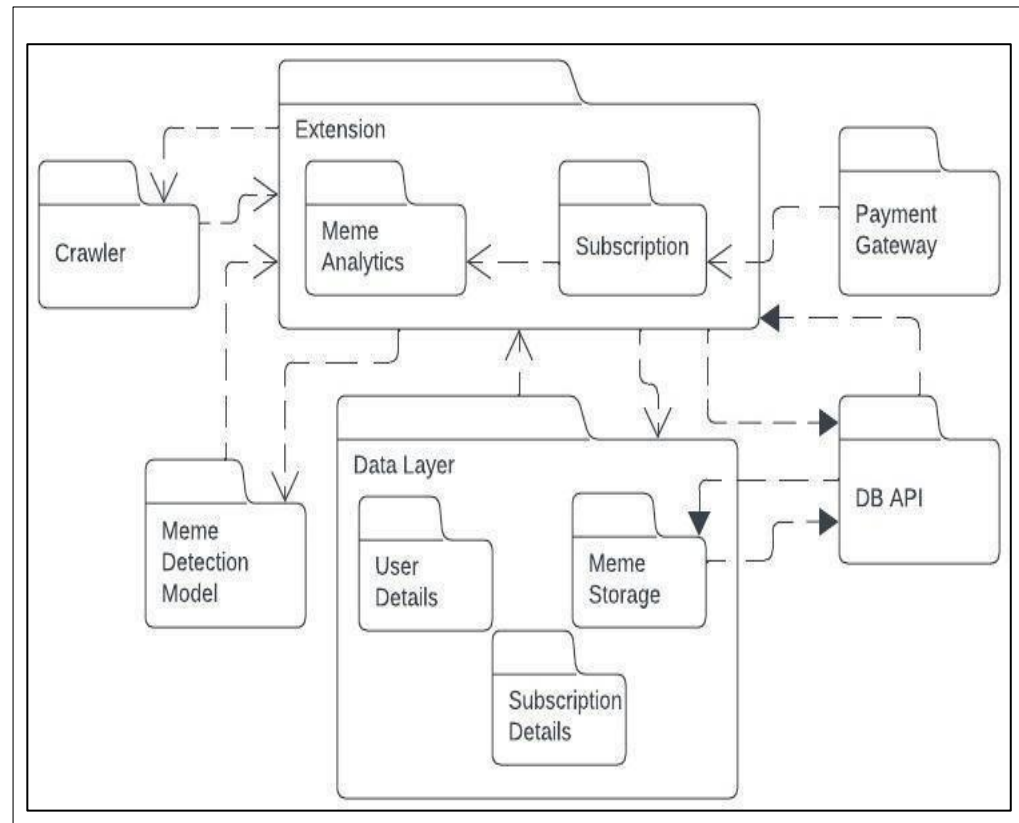


Figure 14: Package Diagram

15. Research

15.1. Introduction

15.1.1. Problem Domain

The rapid spread of hateful content, particularly in the form of memes, presents challenges across social media platforms. This content is complex, often combining text and imagery to convey harmful messages subtly. Current moderation tools struggle with multimodal content, which necessitates a sophisticated model capable of understanding both textual and visual components to detect and mitigate hate speech effectively.

15.1.2. Research Problem Statement

This project focuses on enhancing the detection of hateful memes by improving current multimodal models and developing a Chrome extension to filter this content on social media platforms. By advancing the state-of-the-art models for hate detection in multimodal content, the project aims to create a tool that comprehensively identifies and obscures hateful memes in real-time.

15.2. Literature Review

15.2.1. Hateful Memes Challenge (Kiela et al.)

15.2.1.1. Summary of the Research Item

The Hateful Memes Challenge by Facebook AI introduces a benchmark dataset specifically crafted for the hate speech detection task in multimodal content, combining images and text. This dataset was built to test models on difficult cases that require nuanced understanding, as it includes “benign confounders”—instances that may seem harmless in isolation but are hateful when combined. The challenge aims to push the boundaries of multimodal AI models by presenting a controlled testbed for hate speech detection [1].

15.2.1.2. Critical Analysis (Strengths and Weaknesses)

Strengths: The dataset provides a unique approach by constructing both benign and hateful variants of the same meme, enforcing the need for true multimodal reasoning. It also highlights the limitations of unimodal models in detecting context-dependent hate speech.

Weaknesses: Although comprehensive, the dataset may not cover all cultural or contextual nuances of hate, which could reduce model performance when generalizing beyond the dataset’s scope.

15.2.1.3. Relationship to the Proposed Research Work

This research provides a valuable foundation for developing a Chrome extension that detects hateful memes. The project can leverage this dataset to train and test models, ensuring they are robust in detecting context-specific hate speech across multimodal content.

15.2.2. Are Multimodal Transformers Robust to Missing Modality? (Ma et al.)

15.2.2.1. Summary of the Research Item

This study investigates the robustness of Transformer-based models in multimodal tasks, especially under conditions where one modality (text or image) is missing. It examines various fusion strategies and concludes that Transformers show a notable decline in performance with missing modalities. The study also proposes an automated fusion strategy for optimal results in scenarios with incomplete modality information [4].

15.2.2.2. Critical Analysis (Strengths and Weaknesses)

Strengths: The study highlights the sensitivity of multimodal Transformers to missing data and offers insights into choosing appropriate fusion strategies for different datasets.

Weaknesses: This approach might require significant computational resources for fusion strategy optimization, making real-time detection challenging for resource-constrained applications.

15.2.2.3. Relationship to the Proposed Research Work

This research is relevant for enhancing the robustness of our detection model, particularly for cases where either text or image information may be partially available or ambiguous. Leveraging insights from this study could improve model resilience in real-world settings where memes vary widely in structure.

15.2.3. Modality Invariant Multimodal Learning to Handle Missing Modalities (Saeed et al.)

15.2.3.1. Summary of the Research Item

This research proposes a modality-invariant approach to multimodal learning, enhancing robustness against missing data. By using a single-branch network with weight-sharing across modalities, the model adapts to incomplete multimodal inputs effectively, achieving high performance even with missing information. Extensive evaluations across datasets like Hateful Memes demonstrate the approach's effectiveness compared to traditional multi-branch fusion models [3].

15.2.3.2. Critical Analysis (Strengths and Weaknesses)

Strengths: The single-branch network structure significantly improves efficiency by reducing dependency on complete multimodal data, which is ideal for real-world applications with variable data availability.

Weaknesses: The single-branch approach, while robust, might underperform compared to more complex fusion strategies in cases where both modalities are present and highly informative.

15.2.3.3. Relationship to the Proposed Research Work

The modality-invariant approach aligns well with our project’s goal to create a resilient hate meme detector. Implementing a similar single-branch model could enhance the Chrome extension’s ability to filter content seamlessly, even in cases where meme data is incomplete or ambiguous.

15.2.4. *MUTE: A Multimodal Dataset for Detecting Hateful Memes* (Hossain et al.)

15.2.4.1. Summary of the Research Item

MUTE dataset by Hossain et al. is a multimodal collection focusing on hateful memes in Bengali and code-mixed languages (English-Bengali), addressing a gap in hateful meme detection for resource-constrained languages. This dataset comprises 4,158 memes annotated for hate and non-hate classifications. The study utilizes both unimodal and multi-modal detection methods, including CNN and BiLSTM for individual modalities, and transformer-based fusion for multimodal classification. Experiments reveal that combining textual and visual information improves accuracy by approximately 3% over unimodal models [5].

15.2.4.2. Critical Analysis

Strengths: This dataset highlights the importance of multimodal hate speech detection in low-resource languages. It also addresses the unique challenges of code-mixed language content, which is often overlooked in mainstream datasets and models.

Weaknesses: While the dataset and models provide a substantial contribution, their reliance on specific OCR and transformer-based language models limits flexibility across languages. The dataset also remains small relative to datasets in high-resource languages, potentially affecting model generalization.

15.2.4.3. Relationship to the Proposed Research Work

This dataset serves as a crucial resource for understanding how multimodal detection can be adapted to diverse languages and contexts. Insights gained from the MUTE dataset can help improve our Chrome extension’s effectiveness in identifying hateful content, especially in multilingual and code-mixed contexts.

15.2.5. MOMENTA: A Multimodal Framework for Detecting Harmful Memes and Their Targets (Pramanick et al.)

15.2.5.1. Summary of the Research Item

The MOMENTA framework by Pramanick et al. introduces a multimodal architecture for detecting harmful memes and identifying their targeted entities, based on global and local information from both image and text. MOMENTA extends the HarMeme dataset with additional categories (e.g., US politics) and utilizes various multimodal pre-trained models (e.g., CLIP, VGG-19, and DistilBERT) to improve detection accuracy. This study emphasizes the role of contextual analysis in identifying subtle harmfulness that arises from the combination of text and images [4].

15.2.5.2. Critical Analysis

Strengths: MOMENTA's hierarchical attention mechanism effectively captures both the localized and global contexts, improving its accuracy and interpretability. The extension of the HarMeme dataset with broader contexts adds significant value to hate and harm detection research.

Weaknesses: Despite MOMENTA's robust performance, the reliance on specific pre-trained models (like CLIP) may restrict adaptability in real-time applications or for languages and contexts not covered by these models. Furthermore, the architecture's complexity may limit its utility in lightweight, resource-constrained applications.

15.2.5.3. Relationship to the Proposed Research Work

The MOMENTA framework's emphasis on context-aware multimodal fusion aligns well with our project's objectives to filter out hateful memes accurately. Applying similar hierarchical and cross-modal attention mechanisms could enhance our Chrome extension's capability to understand and block harmful content based on nuanced contextual cues.

15.3. SBNET Diagram

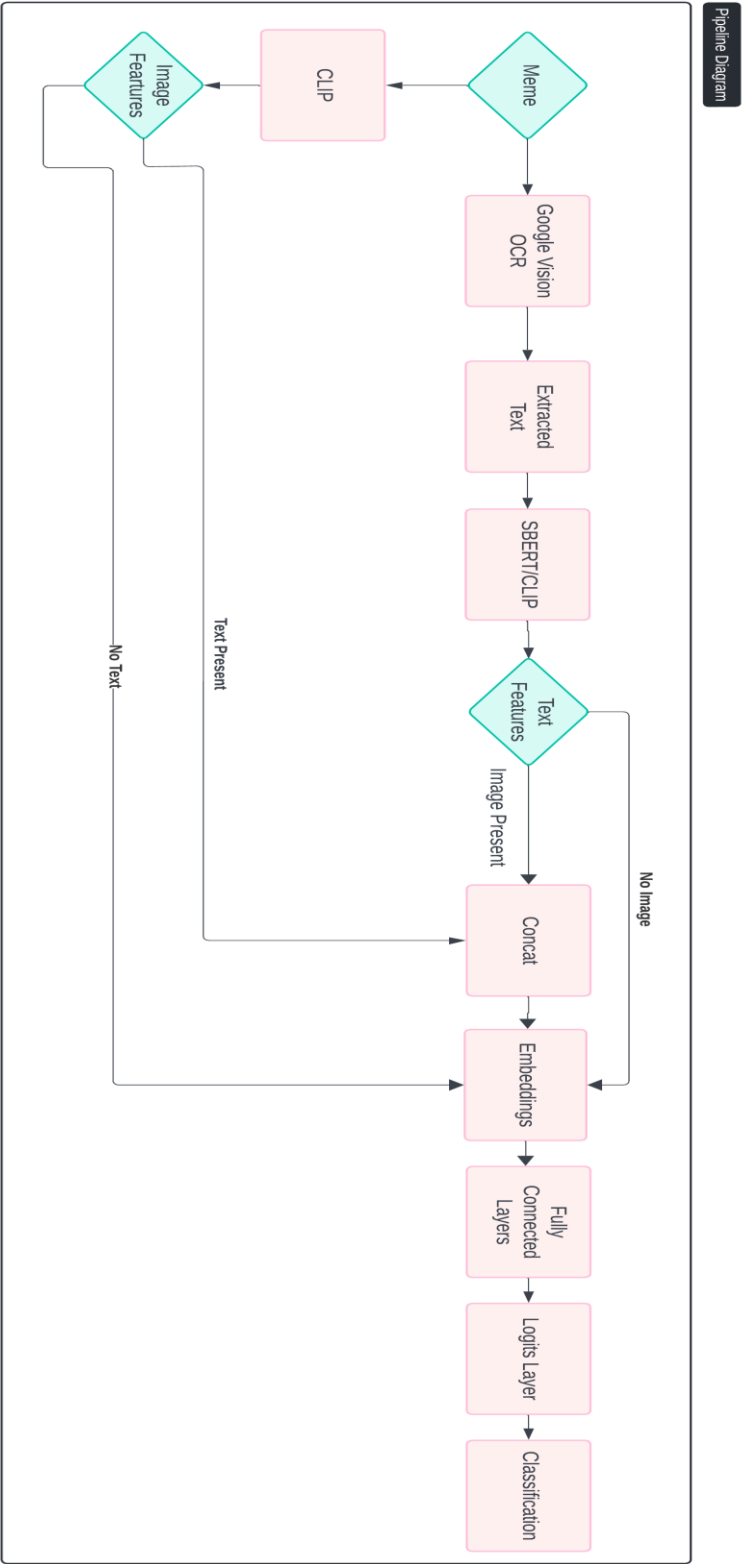


Figure 15: Single-Branch Network

15.4. MBNET Diagram

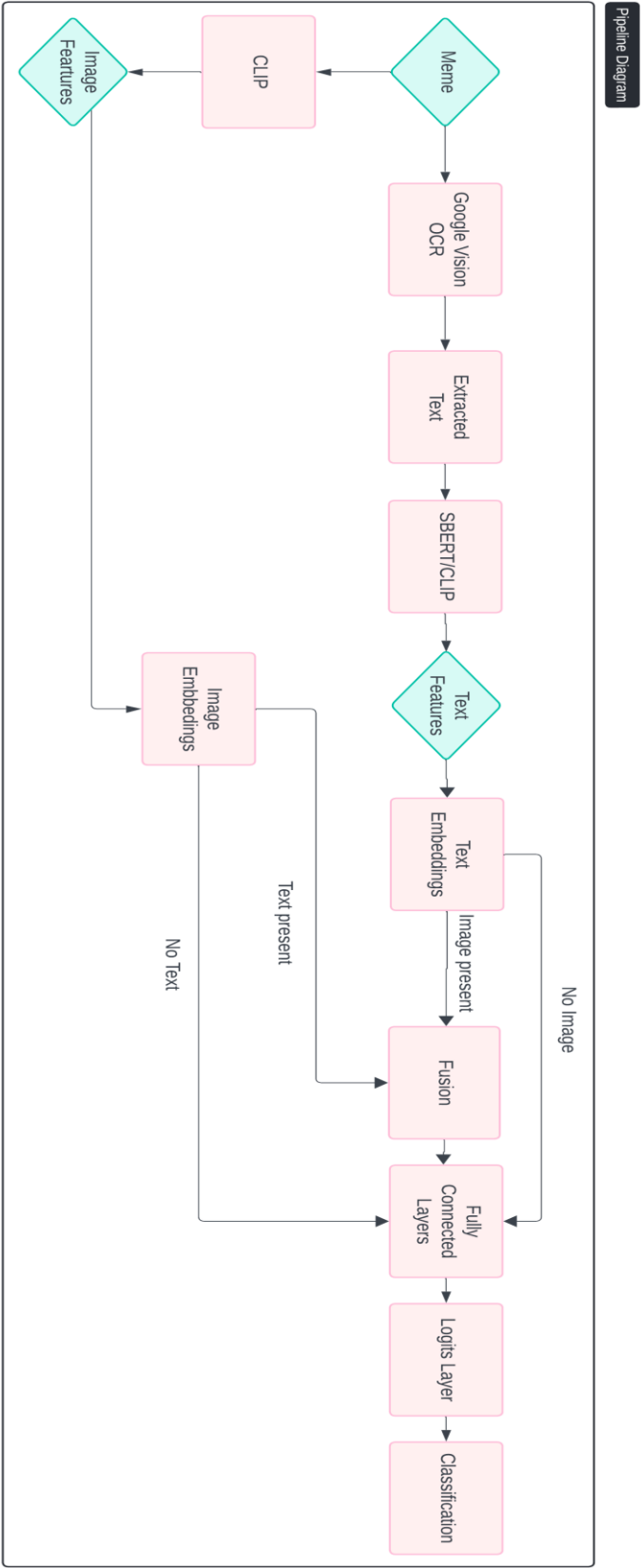


Figure 16: Multi-Branch Network

15.5. Single-Branch Network Evaluation

Text model	Image model	Data	Acc	M-F1	Prec	Rec	AUC	MAE
SBERT	CLIP	HarmC	0.7721	0.7512	0.7557	0.7478	0.7478	0.2279
SBERT	CLIP	HarmP	0.6462	0.6459	0.6469	0.6476	0.6476	0.3538
CLIP	CLIP	HarmC	0.7815	0.7629	0.7655	0.7607	0.7607	0.2185
CLIP	CLIP	HarmP	0.6212	0.6178	0.6185	0.6176	0.6176	0.3788

Table 14: SBNET Evaluation with Complete Modality (768-dim, 2-class, V1)

Text model	Image model	Data	Acc	M-F1	Prec	Rec	AUC	MAE
TEXT	SBERT	HarmC	0.7439	0.7220	0.7242	0.7201	0.7201	0.2561
TEXT	SBERT	HarmP	0.5788	0.5765	0.5765	0.5764	0.5764	0.4212
TEXT	CLIP	HarmC	0.7589	0.7377	0.7408	0.7353	0.7353	0.2411
TEXT	CLIP	HarmP	0.6212	0.6201	0.6201	0.6207	0.6207	0.3788
Image	CLIP	HarmC	0.7815	0.7664	0.7649	0.7682	0.7682	0.2185
Image	CLIP	HarmP	0.6423	0.6357	0.6405	0.6359	0.6359	0.3577

Table 15: SBNET Evaluation with Missing Modality (768-dim, 2-class, V1)

Text model	Image model	Data	Acc	M-F1	Prec	Rec	AUC	MAE
SBERT	CLIP	HarmC	0.7020	0.5178	0.5426	0.5087	0.7640	0.6785
CLIP	CLIP	HarmC	0.7048	0.4956	0.4986	0.4964	0.7454	0.6682
SBERT	CLIP	HarmP	0.5570	0.4996	0.4520	0.5143	0.6726	0.7467
CLIP	CLIP	HarmP	0.5815	0.4682	0.4599	0.4815	0.6747	0.7697

Table 16: SBNET Evaluation with Complete Modality (768-dim, 3-class, V1)

Text model	Image model	Data	Acc	M-F1	Prec	Rec	AUC	MMAE
No Txt	CLIP	HarmC	0.6864	0.5049	0.5175	0.4997	0.7520	0.6901
SBERT	NoImg	HarmC	0.6808	0.4847	0.4909	0.4823	0.7254	0.7060
CLIP	NoImg	HarmC	0.6921	0.5055	0.5151	0.4993	0.7276	0.6879
NoTxt	CLIP	HarmP	0.5628	0.4789	0.4651	0.5017	0.6690	0.7644
SBERT	NoImg	HarmP	0.5440	0.4195	0.4175	0.4225	0.6388	0.8555
CLIP	NoImg	HarmP	0.5296	0.4323	0.4257	0.4459	0.6582	0.8418

Table 17: SBNET Evaluation with Missing Modality (768-dim, 3-class, V1)

15.6. Multi-Branch Model Evaluation

Text model	Image model	Dataset	Acc	M-F1	Prec	Rec	AUC	MMAE	Fusion
SBERT	CLIP	HarmC	0.7909	0.7653	0.7763	0.7583	0.7583	0.2090	Linear
CLIP	CLIP	HarmC	0.8008	0.7695	0.7970	0.7573	0.7583	0.1991	Linear
SBERT	CLIP	HarmC	0.7923	0.7721	0.7745	0.7700	0.7700	0.2076	Gated
CLIP	CLIP	HarmC	0.7923	0.7608	0.7848	0.7498	0.7498	0.2076	Gated
SBERT	CLIP	HarmP	0.6305	0.6305	0.6308	0.6309	0.6309	0.3694	Linear
CLIP	CLIP	HarmP	0.6248	0.6242	0.6243	0.6242	0.6242	0.3751	Linear
SBERT	CLIP	HarmP	0.6291	0.6291	0.6301	0.6299	0.6299	0.3708	Gated
CLIP	CLIP	HarmP	0.6392	0.6391	0.6392	0.6393	0.6393	0.6391	Gated

Table 18: MBNET Evaluation with Complete Modality (768-dim, 2-class, V1)

Text model	Image model	Dataset	Acc	M-F1	Prec	Rec	AUC	MAE	Fusion
Missing	CLIP	HarmC	0.7923	0.7685	0.7766	0.7630	0.7630	0.2076	Linear
Missing	CLIP	HarmC	0.7895	0.7644	0.7741	0.7581	0.7581	0.2104	Gated
SBERT	Missing	HarmC	0.7471	0.7108	0.7275	0.7032	0.7032	0.2528	Linear
SBERT	Missing	HarmC	0.7457	0.7054	0.7280	0.6969	0.6969	0.2542	Gated
CLIP	Missing	HarmC	0.7838	0.7566	0.7687	0.7493	0.7493	0.2161	Linear
CLIP	Missing	HarmC	0.7711	0.7431	0.7533	0.7368	0.7368	0.2288	Gated
Missing	CLIP	HarmP	0.6464	0.6452	0.6462	0.6453	0.6453	0.3535	Linear
Missing	CLIP	HarmP	0.6233	0.6231	0.6252	0.6246	0.6246	0.3766	Gated
SBERT	Missing	HarmP	0.5815	0.5809	0.5810	0.5809	0.5809	0.4184	Linear
SBERT	Missing	HarmP	0.5887	0.5876	0.5924	0.5908	0.5908	0.4112	Gated
CLIP	Missing	HarmP	0.5930	0.5930	0.5935	0.5935	0.5935	0.4069	Linear
CLIP	Missing	HarmP	0.6017	0.6009	0.6009	0.6009	0.6009	0.3982	Gated

Table 19: MBNET Evaluation with Missing Modality (768-dim, 2-class, V1)

Text model	Image model	Dataset	Acc	M-F1	Prec	Rec	AUC	MMAE	Fusion
SBERT	CLIP	HarmC	0.7429	0.4782	0.4655	0.4924	0.7776	0.6817	Linear
CLIP	CLIP	HarmC	0.7316	0.5012	0.5022	0.5037	0.7839	0.6468	Linear
SBERT	CLIP	HarmC	0.7457	0.5287	0.5843	0.5229	0.7917	0.6406	Gated
CLIP	CLIP	HarmC	0.7358	0.5006	0.5049	0.5015	0.7953	0.6528	Gated
SBERT	CLIP	HarmP	0.5988	0.4037	0.3998	0.4106	0.5858	0.8454	Linear
CLIP	CLIP	HarmP	0.6031	0.4063	0.4030	0.4135	0.6032	0.8395	Linear
SBERT	CLIP	HarmP	0.5974	0.4033	0.3983	0.4095	0.5779	0.8475	Gated
CLIP	CLIP	HarmP	0.6017	0.4061	0.4023	0.4123	0.5549	0.8419	Gated

Table 20: MBNET Evaluation with Complete Modality (768-dim, 3-class, V1)

Text model	Image model	Dataset	Acc	M-F1	Prec	Rec	AUC	MAE	Fusion
Missing	CLIP	HarmC	0.7415	0.5159	0.5487	0.5152	0.7817	0.6465	Linear
Missing	CLIP	HarmC	0.7429	0.5235	0.5509	0.5179	0.7817	0.6454	Gated
SBERT	Missing	HarmC	0.7245	0.4614	0.4528	0.4732	0.7814	0.7202	Linear
SBERT	Missing	HarmC	0.7217	0.4580	0.4508	0.4691	0.7802	0.7284	Gated
CLIP	Missing	HarmC	0.7429	0.4783	0.4684	0.4907	0.7684	0.6852	Linear
CLIP	Missing	HarmC	0.75	0.5151	0.6041	0.5063	0.7747	0.6680	Gated
Missing	CLIP	HarmP	0.5815	0.3925	0.3875	0.3987	0.6101	0.8693	Linear
Missing	CLIP	HarmP	0.6002	0.4049	0.4004	0.4115	0.6107	0.8435	Gated
SBERT	Missing	HarmP	0.5613	0.3785	0.3747	0.3848	0.5997	0.8969	Linear
SBERT	Missing	HarmP	0.5627	0.3763	0.3773	0.3859	0.6092	0.8947	Gated
CLIP	Missing	HarmP	0.5598	0.3764	0.3736	0.3839	0.5925	0.8988	Linear
CLIP	Missing	HarmP	0.5656	0.3820	0.3771	0.3878	0.5997	0.8911	Gated

Table 21: MBNET Evaluation with Missing Modality (768-dim, 3-class, V1)

16. References

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